

12.1 Quiz: Complex Numbers — Markscheme

1. [Maximum mark: 5]

AA HL 2023 Nov P1 TZ1 Q07

It is given that $z = 5 + qi$ satisfies $z^2 + iz = -p + 25i$, where $p, q \in \mathbb{R}$. Find p and q .

Solution:

Substitute $z = 5 + qi$: **M1**

$$(5 + qi)^2 + i(5 + qi) = -p + 25i$$

$$25 - q^2 + 10qi + 5i - q = -p + 25i \quad \text{A1}$$

Equate imaginary parts: **M1**

$$10q + 5 = 25$$

$$q = 2 \quad \text{A1}$$

Equate real parts:

$$25 - q^2 - q = -p$$

$$25 - 4 - 2 = -p$$

$$p = -19 \quad \text{A1}$$

2. [Maximum mark: 7]

AA HL 2021 May P2 TZ1 Q08

(a) Find the modulus of zw . [1]

Solution:

$$|zw| = |z||w| = 2 \cdot 8 = 16 \quad \text{A1}$$

(b) Find the argument of zw in terms of k . [2]

Solution:

$$\arg z = \frac{\pi}{5} \text{ and } \arg w = -\frac{2k\pi}{5} \quad \text{M1}$$

$$\arg(zw) = \frac{\pi}{5} - \frac{2k\pi}{5} = \frac{(1 - 2k)\pi}{5} \quad \text{A1}$$

(c) Suppose that $zw \in \mathbb{Z}$. Find the minimum value of k . [3]

Solution:

For zw to be an integer, $\arg(zw)$ must be a multiple of π . **A1**

$$\frac{(1 - 2k)\pi}{5} = m\pi, \text{ where } m \in \mathbb{Z} \quad \text{M1}$$

$$1 - 2k = 5m$$

The least positive integer value is $k = 3$. **A1**

(d) For the value of k found in part (c), find zw . [1]

Solution:

When $k = 3$, $\arg(zw) = -\pi$, so $zw = -16$. **A1**

FT: award A1 for the value of zw obtained from the candidate's value of k in part (c).

3. [Maximum mark: 22]

AA HL 2025 Nov P1 TZ1 Q11

Consider $w = \frac{4i}{1 + i\sqrt{3}}$.

(a) Show that $w = \sqrt{3} + i$. [2]

Solution:

$$\frac{4i}{1 + i\sqrt{3}} \cdot \frac{1 - i\sqrt{3}}{1 - i\sqrt{3}}$$

M1

$$= \frac{4\sqrt{3} + 4i}{4} = \sqrt{3} + i$$

A1 AG

(b)(i) Find $|w|$. [2]

Solution:

$$|w| = \sqrt{(\sqrt{3})^2 + 1^2}$$

M1

$$|w| = 2$$

A1

(b)(ii) Find $\arg w$. [1]

Solution:

$$\arg w = \frac{\pi}{6}$$

A1

(c) Use de Moivre's theorem to find the two square roots of w . [4]

Solution:

$$w = 2e^{i\pi/6}, \text{ so one square root is}$$

M1

$$\sqrt{2}e^{i\pi/12}$$

A1

The other root is obtained by adding or subtracting π to the argument. **M1**

$$\sqrt{2}e^{-11\pi i/12}$$

A1

Accept $\sqrt{2}e^{13\pi i/12}$ if the requested argument range is not enforced.

(d)(i) Show that $a^2 = \frac{\sqrt{3}}{2} + 1$. [6]

Solution:

$$(a + bi)^2 = a^2 - b^2 + 2abi = \sqrt{3} + i \quad \text{M1}$$

$$a^2 - b^2 = \sqrt{3} \text{ and } 2ab = 1 \quad \text{A1}$$

$$b = \frac{1}{2a}, \text{ so } a^2 - \left(\frac{1}{2a}\right)^2 = \sqrt{3} \quad \text{M1}$$

$$4a^4 - 4\sqrt{3}a^2 - 1 = 0 \quad \text{A1}$$

$$\text{Solving as a quadratic in } a^2 \text{ gives } a^2 = \frac{\sqrt{3}}{2} \pm 1. \quad \text{M1}$$

$$\text{Since } a^2 > 0, a^2 = \frac{\sqrt{3}}{2} + 1. \quad \text{R1 AG}$$

(d)(ii) Find the value of b^2 . [2]

Solution:

$$b^2 = a^2 - \sqrt{3} \quad \text{M1}$$

$$b^2 = 1 - \frac{\sqrt{3}}{2} \quad \text{A1}$$

FT: award A1 for b^2 correctly found from the candidate's value of a^2 in part (d)(i).

(e) Hence express $\tan \frac{\pi}{12}$ in the form $p + q\sqrt{3}$. [5]

Solution:

$$a + bi \text{ is a square root of } w \text{ with argument } \frac{\pi}{12}, \text{ so } \tan \frac{\pi}{12} = \frac{b}{a} \quad \text{M1}$$

$$\tan \frac{\pi}{12} = \sqrt{\frac{b^2}{a^2}} \quad \text{A1}$$

$$\text{Substitute the values of } a^2 \text{ and } b^2: \quad \text{M1}$$

$$\tan \frac{\pi}{12} = \sqrt{\frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}}} = \sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}}} \quad \text{A1}$$

$$\tan \frac{\pi}{12} = 2 - \sqrt{3} \quad \text{A1}$$

FT: award the substitution marks for consistent use of the candidate's values from part (d).

4. [Maximum mark: 20]

AA HL 2024 Nov P1 TZ0 Q12

(a) Find z_1, z_2, z_3 and z_4 . [6]**Solution:**

$$|16i| = 16 \text{ and } \arg(16i) = \frac{\pi}{2} \quad \text{A1}$$

Use de Moivre's theorem. M1

$$z_1 = 2 \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right) \quad \text{A1}$$

Other roots are found by adding $\frac{\pi}{2}$ to the argument. M1

$$z_2 = 2 \left(\cos \frac{5\pi}{8} + i \sin \frac{5\pi}{8} \right) \quad \text{A1}$$

$$z_3 = 2 \left(\cos \frac{9\pi}{8} + i \sin \frac{9\pi}{8} \right), z_4 = 2 \left(\cos \frac{13\pi}{8} + i \sin \frac{13\pi}{8} \right) \quad \text{A1}$$

(b) Find the common ratio in Cartesian form. [3]

Solution:

$$\frac{z_2}{z_1} \quad \text{M1}$$

$$= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \quad \text{A1}$$

$$= i \quad \text{A1}$$

FT: award marks for the ratio computed from the candidate's roots in part (a).

(c) Plot A, B, C and D on an Argand diagram. [3]

Solution:

Point A approximately in the first quadrant. A1

All four points approximately the same distance from the origin. A1

Approximate angular separation of $\frac{\pi}{2}$. A1(d) Determine a and b for the conjugate-root equation. [3]**Solution:**

$$\text{For each root, } (z_j^*)^4 = (z_j^4)^* \quad \text{A1}$$

$$= (16i)^* \quad \text{A1}$$

$$\text{Therefore } v^4 = -16i, \text{ so } a = 0 \text{ and } b = -16. \quad \text{A1}$$

(e) Find the least possible value of p and the corresponding value of q . [5]

Solution:

The first midpoint has argument $\frac{\pi}{8} + \frac{\pi}{8} = \frac{3\pi}{8}$. **A1**

The modulus of each midpoint is $\sqrt{2}$. **A1**

For $(\sqrt{2}e^{3\pi i/8})^p$ to be positive real, the argument must be a multiple of 2π . **M1**

The least possible value is $p = 16$. **A1**

Then $(\sqrt{2})^{16} = 2^8$, so $q = 8$. **A1**

5. [Maximum mark: 22]

AA HL 2023 May P1 TZ2 Q11

(a) Show that $u = 2e^{i\frac{2\pi}{3}}$. [3]

Solution:

$$|u| = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2 \quad \text{A1}$$

Reference angle $\frac{\pi}{3}$, with u in quadrant II. **M1**

$$\arg u = \frac{2\pi}{3}, \text{ so } u = 2e^{i\frac{2\pi}{3}}. \quad \text{A1 AG}$$

(b)(i) Find the smallest positive integer n such that u^n is real. [3]

Solution:

$$u^n = 2^n e^{i\frac{2n\pi}{3}} \text{ is real when } \sin \frac{2n\pi}{3} = 0. \quad \text{M1}$$

$$\text{Thus } \frac{2n\pi}{3} = k\pi, \text{ where } k \in \mathbb{Z}. \quad \text{A1}$$

The smallest positive integer is $n = 3$. **A1**

(b)(ii) Find the value of u^n for this value of n . [2]

Solution:

$$u^3 = 2^3 e^{2\pi i} \quad \text{M1}$$

$$u^3 = 8 \quad \text{A1}$$

FT: award marks for substituting the candidate's multiple-of-3 value from part (b)(i).

(c)(i) Given that u is a root, find the other roots. [5]

Solution:

Since coefficients are real, $-1 - \sqrt{3}i$ is also a root. **A1**

Let the third root be c . Use the sum of roots. **M1**

$$(-1 + \sqrt{3}i) + (-1 - \sqrt{3}i) + c = -5 \quad \mathbf{A1}$$

$$-2 + c = -5 \quad \mathbf{A1}$$

$$c = -3, \text{ so the other roots are } -1 - \sqrt{3}i \text{ and } -3. \quad \mathbf{A1}$$

(c)(ii) Find the roots of $1 + 5w + 10w^2 + 12w^3 = 0$. [4]

Solution:

Compare with $z^3 + 5z^2 + 10z + 12 = 0$ using $z = \frac{1}{w}$. **A1A1**

Thus $w = \frac{1}{z}$ for each root. **M1**

$$w = -\frac{1}{3} \text{ and } w = \frac{1}{-1 \pm \sqrt{3}i} = \frac{-1 \mp \sqrt{3}i}{4}. \quad \mathbf{A1}$$

Accept $-\frac{1}{3}, \frac{-1+\sqrt{3}i}{4}$ and $\frac{-1-\sqrt{3}i}{4}$.

(d) Solve $z^2 = 2z^*, z \neq 0$. [5]

Solution:

Let $z = a + bi$. Then $(a + bi)^2 = 2(a - bi)$. **A1**

Equate real and imaginary parts: $a^2 - b^2 = 2a$ and $2ab = -2b$. **M1**

$2b(a + 1) = 0$ **M1**

If $b = 0$, then $a^2 = 2a$, so $a = 2$ since $z \neq 0$. **A1**

If $a = -1$, then $1 - b^2 = -2$, so $b = \pm\sqrt{3}$. The roots are 2 and $-1 \pm \sqrt{3}i$. **A1**

6. [Maximum mark: 18]

AA HL 2022 May P1 TZ2 Q12

(a) Show that $z_1 z_2^* = r_1 r_2 e^{i(\alpha - \theta)}$. [2]

Solution:

$$z_2^* = r_2 e^{-i\theta} \quad \mathbf{A1}$$

$$z_1 z_2^* = r_1 e^{i\alpha} r_2 e^{-i\theta} = r_1 r_2 e^{i(\alpha - \theta)} \quad \mathbf{A1 \ AG}$$

(b) Show that $Z_1 O Z_2$ is right-angled. [2]

Solution:

$$\operatorname{Re}(z_1 z_2^*) = r_1 r_2 \cos(\alpha - \theta) = 0 \quad \mathbf{A1}$$

Since $r_1, r_2 > 0$ and $0 < \alpha - \theta < \pi$, $\alpha - \theta = \frac{\pi}{2}$, so the triangle is right-angled. **A1 \ AG**

(c)(i) Express z_1 in terms of z_2 . [2]

Solution:

For an equilateral triangle, $r_1 = r_2$ and $\alpha - \theta = \frac{\pi}{3}$. **M1**

$$z_1 = z_2 e^{i\pi/3} \quad \text{A1}$$

(c)(ii) Hence show that $z_1^2 + z_2^2 = z_1 z_2$. [4]

Solution:

Substitute $z_1 = z_2 e^{i\pi/3}$ into the left-hand side. **M1**

$$z_1^2 + z_2^2 = z_2^2 e^{2\pi i/3} + z_2^2 = z_2^2 (e^{2\pi i/3} + 1) \quad \text{A1}$$

$$e^{2\pi i/3} + 1 = e^{\pi i/3} \quad \text{A1}$$

$$z_1^2 + z_2^2 = z_2^2 e^{\pi i/3} = z_2 (z_2 e^{\pi i/3}) = z_1 z_2 \quad \text{A1 AG}$$

(d) Use part (c)(ii) to show that $a^2 - 3b = 0$. [5]

Solution:

For roots z_1 and z_2 , $z_1 + z_2 = -a$ and $z_1 z_2 = b$. **A1**

$$a^2 = (z_1 + z_2)^2 \quad \text{A1}$$

$$a^2 = z_1^2 + z_2^2 + 2z_1 z_2 \quad \text{A1}$$

Using part (c)(ii), $a^2 = z_1 z_2 + 2z_1 z_2$. **M1**

$$a^2 = 3b, \text{ so } a^2 - 3b = 0. \quad \text{A1 AG}$$

(e) Deduce that only one equilateral triangle can be formed. [3]

Solution:

With $b = 12$, $a^2 - 3(12) = 0$, so $a = \pm 6$. **A1**

For $a = -6$, the roots are $3 \pm \sqrt{3}i$, which do not satisfy $0 < \alpha - \theta < \pi$ with the required orientation. **R1**

For $a = 6$, the roots are $-3 \pm \sqrt{3}i$ and $\alpha - \theta = \frac{\pi}{3}$, so only one equilateral triangle is possible. **A1 AG**